CNSCC.160 – Fundamentals of Communication Systems

Tutorial 1

Question 1:

What are the various transducers used in communication systems?

A transducer is a device that converts energy from one form to another. In communication, they usually convert physical signals into electrical signals (and vice versa).

- **Input Transducers:** Convert physical quantities (sound, light, pressure) into electrical signals. *Examples: Microphones, cameras, sensors.*
- **Output Transducers:** Convert electrical signals back into a physical form. *Examples: Loudspeakers, LED displays, actuators.*

Question 2:

What do you understand by the term baseband signal?

A **baseband signal** is the original information signal as it is produced by the source, without any modulation. It contains frequencies that are very near to zero (DC). In communication systems, baseband signals are often "modulated" onto a higher frequency carrier wave to be transmitted over long distances.

Question 3:

Briefly describe the source coding.

Source coding is the process of encoding information from a source to make it more efficient for transmission or storage. Its primary goal is **data compression** by removing redundancy.

Question 4:

Characterise the main areas of image processing: image compression, image enhancement and image understanding. State what their goals are and give examples.

Area	Goal	Example
Image Compression	To reduce the storage space or bandwidth required to represent an image.	Converting a raw photo into a **JPEG** file.
Image Enhancement	To improve the visual appearance of an image or provide a "better" transform for analysis.	**Adjusting contrast** or sharpening a blurred photo.
Image Understanding	To extract high-level information from an image so a system can "interpret" what it sees.	**Object recognition** (e.g., identifying a car in a traffic feed).

Question 5:

Explain the difference between the high- and low-level image processing tasks.

- Low-Level Tasks: Primitive operations where input and output are images. Examples include noise reduction and contrast enhancement.

- High-Level Tasks: Image analysis, computer vision. The input is an image, but the output is often a description or a decision. Examples include scene analysis and autonomous navigation.

Question 6:

Explain the digital image representation.

A digital image is represented as a 2D array (matrix) of **pixels** (values).

- Each pixel has a **coordinate** (x, y) and an **intensity value** (grey level or colour).
- For a standard 8-bit grayscale image, intensity values range from 0 (black) to 255 (white).
- Colour images are typically represented by three such matrices (Red, Green, and Blue).

Question 7:

Compare the principle of work of the human eye with the camera.

Both systems capture light and focus it on a light-sensitive surface, but they differ in mechanics:

- **Focusing:** The eye changes the **shape** of the lens, while a camera lens moves **physically closer or further** from the sensor.
- **Sensing:** The eye uses **photoreceptors** (rods and cones) on the retina; a camera uses a **CCD or CMOS sensor**.
- **Processing:** The eye sends signals to the brain for interpretation; a camera saves data to a storage medium or digital processor.

Question 8:

Explain the principle of work of the optical camera compared with the infrared (thermal) camera.

- Optical Camera: Detects reflected visible light (0.4 to 0.7 microns). It requires an external light source (like the sun or a bulb) to see.

- Infrared Camera: Detects emitted heat (thermal radiation) from objects. It does not require visible light and can "see" in total darkness by capturing the infrared energy emitted by any object with a temperature above absolute zero.

Question 9:

Why are the infrared (IR) rays able to penetrate given medium rather easily?

Infrared rays have **longer wavelengths** than visible light. Because of this:

1. **Lower Scattering:** Longer wavelengths are scattered less by small particles in the atmosphere (like smoke, dust, or light fog).
2. **Material Interaction:** Many materials that are opaque to visible light (like certain plastics or silicon) are transparent to specific IR wavelengths, allowing the rays to pass through rather than being absorbed or reflected.